

UNIQUENESS OF THE RENORMALIZED QUARTIC-OSCILLATOR CRITICAL POINT

ZHIYUAN YAO

ABSTRACT. For the ground state G_b of $h_b = \frac{1}{2}(-d^2/dx^2 + bx^2) + x^4$, define $m(b) = b + 12\langle G_b, x^2 G_b \rangle$. We prove that m' has exactly one real zero b_* . Moreover, $-16/5 < b_* < 0$ and $m''(b_*) > 0$. Exact dilation reduces the problem to the level equation $\lambda_1''(\alpha) = 4/3$ for the first Montgomery eigenvalue. Analytic estimates confine every target to $0 < \alpha < 4/5$. At any target, a reduced resolvent identity turns an even-sector gap bound into strict downward crossing. The required gap bound is certified by exact moment recurrences and an exhaustive rational Bernstein subdivision of a three-dimensional box. A translated Gaussian forces existence, and a zero-topology lemma then gives uniqueness. All load-bearing computer-assisted inequalities use exact rational arithmetic.

1. INTRODUCTION

The renormalized oscillator coefficient m was introduced by Baez and Zhou, who conjectured from finite-difference evidence that it has one critical point [2]. Its quartic spectral background includes the classical perturbative and analytic work of Bender–Wu and Simon [4, 18]. Analytic perturbation theory, the standard theory of discrete Schrödinger spectra, and exponential decay of bound states provide the operator-theoretic setting used below [12, 16, 1].

Under the exact dilation used below, the spectral problem becomes the Montgomery family

$$H_\alpha = -\frac{d^2}{dt^2} + \left(\frac{t^2}{2} - \alpha\right)^2,$$

which has a substantial magnetic-spectral history beginning with Montgomery and continuing through magnetic-well and surface-superconductivity analysis [13, 11, 15, 7, 6]. Helffer revisited the $k = 1$ model, and Helffer–Persson treated the associated higher-order anharmonic family [9, 8]. Recent work connects quartic spectral summability to the Engel group [3], and the anharmonic oscillator itself remains a live subject [19]. Helffer and Léautaud proved uniqueness of the zero of $\lambda_1'(\alpha)$ [10]; that theorem does not decide the present problem, whose exact stationarity condition is the second-derivative level equation $\lambda_1''(\alpha) = 4/3$.

Our proof separates global analysis from a finite exact certificate. First, all solutions of the level equation are placed in a compact rational interval. Second, reduced-resolvent identities show that strict transversality follows from an even-gap inequality. Moment and hypervirial constraints reduce failure of that inequality to the simultaneous nonnegativity of nine rational polynomials on a compact box. The Bernstein basis and its subdivision properties are surveyed in [5]; interval and verification methods provide the wider validated-computation context [14, 17]. Here every Bernstein coefficient and every de Casteljau subdivision is instead computed in exact rational arithmetic. The resulting exhaustive tree excludes the box in 757 nodes, so the certificate is part of the proof rather than a floating-point experiment.

Theorem 1.1. *Let h_b be the standard form-sum quartic oscillator on $L^2(\mathbb{R})$, let G_b be its normalized positive ground state, and define*

$$m(b) = b + 12\langle G_b, x^2 G_b \rangle.$$

Then m has exactly one real critical point. It lies in $(-16/5, 0)$ and is a strict local minimum.

Remark 1.2. Only the rational Bernstein exclusion in the transversality argument is computer assisted. The target localization, existence argument, and zero-counting step are analytic. All nine polynomials, their affine cube map, the subdivision rule, the strict terminal test, and the exact coverage argument are stated below.

2. REDUCTION TO A CURVATURE LEVEL SET

Proposition 2.1. *Let $E(b)$ be the ground eigenvalue of h_b and let*

$$H_\alpha = -\frac{d^2}{dt^2} + \left(\frac{t^2}{2} - \alpha\right)^2, \quad \lambda(\alpha) = \inf \text{spec } H_\alpha. \quad (2.1)$$

Then λ is real-analytic on $\mathbb{R}$, and with

$$f(\alpha) := \lambda''(\alpha) - \frac{4}{3} \quad (2.2)$$

one has

$$m'(b) = \frac{3}{2} f(-b/4). \quad (2.3)$$

Consequently m has exactly one critical point if and only if f has exactly one real zero.

Proof. The Feynman–Hellmann formula applied to $\partial_b h_b = x^2/2$ gives

$$m(b) = b + 24E'(b), \quad m'(b) = 1 + 24E''(b). \quad (2.4)$$

The unitary dilation $t = \sqrt{2}x$ together with $\alpha = -b/4$ carries h_b onto $H_\alpha - \alpha^2$, so $E(b) = \lambda(-b/4) - b^2/16$.

For the analyticity, write $H_\alpha = p^2 + t^4/4 - \alpha t^2 + \alpha^2$. Multiplication by t^2 is infinitesimally form-bounded with respect to $p^2 + t^4/4$, so the forms are closed on the common domain

$$\mathcal{Q} = H^1(\mathbb{R}) \cap L^2(\mathbb{R}, t^4 dt),$$

and $\{H_\alpha\}$ is an analytic family of type (B) with compact resolvent. In one dimension the ground eigenvalue is simple by Sturm–Liouville theory, so the local analytic perturbation branches concatenate into a real-analytic λ on all of $\mathbb{R}$; in particular f is C^1 .

Substituting $E(b) = \lambda(-b/4) - b^2/16$ into (2.4),

$$m'(b) = -2 + \frac{3}{2} \lambda''(-b/4) = \frac{3}{2} f(-b/4),$$

which is (2.3). Since $b \mapsto -b/4$ is a bijection of $\mathbb{R}$, the zeros of m' and of f correspond one to one. $\square$

Call α a *target* when $f(\alpha) = 0$. The rest of the paper shows that there is exactly one target, and that it lies in $(0, 4/5)$.

3. EVERY TARGET LIES IN $(0, 4/5)$

3.1. The nonpositive half-line.

Proposition 3.1. *$f(\alpha) > 0$ for every $\alpha \leq 0$. In particular no target is nonpositive.*

Proof. Put $\alpha = -s$ with $s \geq 0$, remove the additive constant s^2 , and set

$$K_s = p^2 + s t^2 + \frac{t^4}{4}, \quad e_j(s) = \lambda_j(-s) - s^2.$$

Let u be the normalized ground state of K_s and put

$$a = \langle t^2 \rangle, \quad c = \langle t^4 \rangle, \quad v = c - a^2, \quad g = (t^2 - a)u,$$

with R the positive reduced resolvent above e_1 . Differentiating the eigenvalue equation twice gives

$$\lambda''(-s) = 2 - 2S, \quad S = \langle g, Rg \rangle. \quad (3.1)$$

Only even excited states contribute to S , so with $\Delta = e_3 - e_1$ the spectral theorem gives $S \leq v/\Delta$. The virial identity $2\langle p^2 \rangle = 2sa + c$ yields

$$e_1 = 2sa + \frac{3}{4}c \geq \frac{3}{4}c, \quad v \leq c \leq \frac{4}{3}e_1. \quad (3.2)$$

It remains to separate e_3 from e_1 . For $d > 0$ let $r_d(s) > 0$ solve

$$s = r_d^2 - \frac{d}{4r_d}, \quad F_d(s) = r_d - \frac{d}{16r_d^2}. \quad (3.3)$$

A centred Gaussian of width ρ has Rayleigh quotient $\rho/2 + s/(2\rho) + 3/(16\rho^2)$, whose stationarity equation is (3.3) with $d = 3$; hence $e_1 \leq F_3$. For every $\gamma > s$,

$$\frac{t^4}{4} + st^2 \geq \gamma t^2 - (\gamma - s)^2,$$

and comparison with $p^2 + \gamma t^2$, optimized at $\sqrt{\gamma} = r_5(s)$, gives $e_3 \geq 5F_5$. At fixed s , implicit differentiation of (3.3) gives

$$\frac{\partial F_d(s)}{\partial d} = \frac{1}{16r_d(s)^2} > 0, \quad (3.4)$$

so $e_3 \geq 5F_5 > 5F_3 \geq 5e_1$ and therefore $\Delta > 4e_1$. With (3.1) and (3.2),

$$S < \frac{(4/3)e_1}{4e_1} = \frac{1}{3}, \quad \text{hence} \quad f(\alpha) = \lambda''(\alpha) - \frac{4}{3} > 0. \quad (3.5)$$

□

3.2. Necessary inequalities at a target.

Lemma 3.2. *Let α be a target, let u be the corresponding normalized ground state, and write $a = \langle t^2 \rangle$, $v = \langle t^4 \rangle - a^2$. Then*

$$v^2 \leq \frac{4}{3}a, \quad \text{so } v \leq U(a) := 2\sqrt{\frac{a}{3}}, \quad (3.6)$$

$$3a - 2\alpha > 0, \quad (3.7)$$

$$v \geq L(a, \alpha) := 2\alpha a - a^2 + \frac{1}{2a}, \quad (3.8)$$

$$P(a, \alpha, v) := v \left[3a^3 - 6\alpha a^2 + (3a + 2\alpha)v + \frac{9}{2} \right] - 10a^2 \geq 0. \quad (3.9)$$

Proof. Inequality (3.6) is the inverse- and energy-weighted response identity at a target; (3.7) is the momentum double commutator; (3.8) combines the virial identity with the Heisenberg inequality; and (3.9) is Robertson's inequality for the pair t^2 and $D = (tp + pt)/2$, followed by the t^2 hypervirial identity. □

Lemma 3.3. $\lambda'(\alpha) = 2\alpha - a > 0$ for every $\alpha \geq 1$; in particular $\alpha > a/2$ there.

Proof. At a positive critical point of λ , half-line integration by parts gives

$$(\lambda - \alpha^2) u(0)^2 = \int_0^\infty (t + \sqrt{2\alpha})(t - \sqrt{2\alpha})^2 u(t)^2 dt > 0. \quad (3.10)$$

A centred Gaussian of width $\rho = (2\alpha)^{-1}$ gives $\lambda(\alpha) \leq 1/(4\alpha) + \frac{3}{4}\alpha^2 \leq \alpha^2$ for $\alpha \geq 1$, contradicting (3.10). So λ' has no zero on $[1, \infty)$, and the well-bracketing asymptotic $\lambda(\alpha) \sim \sqrt{2\alpha}$ selects the positive sign. The half-line identity (3.10) is due to Pan and Kwek [15] for the first eigenvalue, and the proof recalled here is that of Helffer and Persson Sundqvist [8]; the comparison argument, the

sign selection and the asymptotic are taken from Helffer and Léautaud [10] and are not derived here. $\square$

Proposition 3.4. *Every target satisfies $\alpha < 1$.*

Proof. Suppose a target had $\alpha \geq 1$. Then (3.7) and Lemma 3.3 give $a > 2/3$, so $U(a) < 3a^2$ and P is decreasing in α . Using (3.6), (3.9) and Lemma 3.3,

$$10a^2 < U(a) \left(4aU(a) + \frac{9}{2} \right) = \frac{16}{3}a^2 + 9\sqrt{\frac{a}{3}},$$

whence

$$a < a_0 := \left(\frac{9\sqrt{3}}{14} \right)^{2/3} < \frac{9}{8}. \quad (3.11)$$

On the other hand (3.6) and (3.8) give

$$\alpha \leq F(a) := \frac{a}{2} + \frac{1}{\sqrt{3a}} - \frac{1}{4a^2}. \quad (3.12)$$

Writing $y = a^{3/2}$,

$$2F'(a) = \frac{(y - 1/(2\sqrt{3}))^2 + 11/12}{y^2} > 0,$$

so F is increasing, and

$$F(9/8) = \frac{473}{1296} + \sqrt{\frac{8}{27}} < \frac{2}{5} + \frac{3}{5} = 1.$$

Together with (3.11) this contradicts $\alpha \geq 1$. $\square$

3.3. Sharpening the endpoint to $4/5$.

Lemma 3.5. *At a target, a unit-width Gaussian and the virial identity give*

$$v \leq E(a, \alpha) := -a^2 + \frac{8}{3}\alpha a - \frac{2}{3}\alpha + \frac{11}{12}, \quad (3.13)$$

and for $0 < \alpha < 1$ the polynomial P of (3.9) is strictly increasing in $v > 0$, because

$$3a^3 - 6\alpha a^2 + \frac{9}{2} > \frac{17}{18} + \frac{(3a - 4)^2(3a + 2)}{9} > 0. \quad (3.14)$$

Proposition 3.6. *Every target satisfies $0 < \alpha < 4/5$.*

Proof. By Propositions 3.1 and 3.4 a target lies in $(0, 1)$; suppose one had $\alpha \geq 4/5$. From $L \leq U$ one gets $\alpha \leq F(a)$ with F as in (3.12), and F is increasing with

$$F(19/20) = \frac{19}{40} + \sqrt{\frac{20}{57}} - \frac{100}{361} < \frac{4}{5},$$

the radical comparison following exactly from $(8693/14440)^2 - 20/57 = 7216747/625540800 > 0$. Hence $a > 19/20$, so $U(a) < 3a^2$ and $\partial_\alpha P = 2v(v - 3a^2) < 0$.

Case $19/20 < a \leq 2$. Put $x = \sqrt{a/3}$, so $9/16 < x < 5/6$. Monotonicity in v (Lemma 3.5) and in α gives $P(a, \alpha, v) \leq P(a, 4/5, 2x) = \frac{x}{5}Q(x)$ with

$$Q(x) = 810x^6 - 432x^4 - 270x^3 + 32x + 45. \quad (3.15)$$

The Sturm chain of Q has three sign variations at both $9/16$ and $5/6$, so Q has no zero between them, and $Q(3/4) = -76659/2048 < 0$. Thus $Q < 0$ on the whole range, contradicting $P \geq 0$.

Case $a \geq 2$. Positivity of v together with (3.13) and $\alpha < 1$ confines this case to $2 \leq a < 3$. Since P increases in v ,

$$P(a, \alpha, v) \leq R(a, \alpha) := P(a, \alpha, E(a, \alpha)).$$

Exact differentiation gives

$$\partial_\alpha^2 R = \frac{4(4a-1)}{9} J(a, \alpha), \quad J = -6a^2 + 48a\alpha - 6a - 12\alpha + 11. \quad (3.16)$$

On $2 \leq a \leq 3$, $4/5 \leq \alpha \leq 1$ one has $J > 0$: it increases in α , and the concave quadratic $J(a, 4/5)$ has endpoint values $211/5$ and $223/5$. Also

$$\partial_\alpha R(a, 4/5) = \frac{S(a)}{1800}, \quad S = -11760a^3 + 47532a^2 + 22424a - 6343,$$

and S is concave on $[2, 3]$ with positive endpoint values 134553 and 171197 , so R increases in α . Finally

$$R(a, 1) = -\frac{K(a)}{144}, \quad K = 876a^3 - 176a^2 - 2139a - 180,$$

with $K(2) = 1846$, $K'(2) = 7669$ and $K''(a) = 5256a - 352 > 0$ for $a \geq 2$, so $K > 0$ and $R(a, \alpha) < R(a, 1) < 0$, again contradicting $P \geq 0$. $\square$

4. STRICT TRANSVERSALITY AT EVERY TARGET

4.1. A sufficient even-gap criterion.

Lemma 4.1. *Let α be a target with normalized ground state u , and set $a = \langle t^2 \rangle$, $g = (t^2 - a)u$, $R = Q(H_\alpha - \lambda_1)^{-1}Q$, $\phi = Rg$, and $\Delta = \lambda_3 - \lambda_1$ for the first even gap. Then*

$$\langle g, Rg \rangle = \frac{1}{3}, \quad \lambda'''(\alpha) = -6\langle \phi, (t^2 - a)\phi \rangle, \quad (4.1)$$

and

$$\Delta > 3a^2 \implies \lambda'''(\alpha) < 0. \quad (4.2)$$

Proof. Differentiating in the parallel-transport gauge gives (4.1); the first identity is the target condition $\lambda'' = 4/3$ rewritten through (3.1). The minimax principle and the Cauchy-Schwarz inequality give respectively

$$\|\phi\|^2 \leq \frac{1}{3\Delta}, \quad \langle \phi, t^2 \phi \rangle \geq \frac{1}{9a}, \quad (4.3)$$

so that $\langle \phi, (t^2 - a)\phi \rangle \geq (\Delta - 3a^2)/(9a\Delta)$, which is positive exactly when $\Delta > 3a^2$. $\square$

4.2. Isolating the third pure-quartic eigenvalue.

Lemma 4.2. *Let $L = p^2 + t^4/4$. Then $\lambda_3(L) > 467/100$.*

Proof. Take the even trial vector $\psi(t) = P(t^2)e^{-t^2/2}$ with the following decimal strings read as exact rationals:

$$\begin{aligned} P(y) = & 1 - 1.8476436929042896 y - 0.13114586665910161 y^2 + 0.050796559364988544 y^3 \\ & + 0.0027570484421582321 y^4 - 0.0012003885725975921 y^5 \\ & + 0.000098882369961766198 y^6 - 0.0000033908436935998999 y^7 \\ & + 0.000000042707174142248135 y^8. \end{aligned} \quad (4.4)$$

With $\mu = \langle \psi, L\psi \rangle / \|\psi\|^2$ and $r^2 = \|(L - \mu)\psi\|^2 / \|\psi\|^2$, Gaussian moments reduce both to exact rationals, and direct reduction verifies

$$\frac{467}{100} < \mu < \frac{88439}{10000}, \quad \left(\mu - \frac{467}{100}\right)^2 > r^2, \quad \left(\frac{88439}{10000} - \mu\right)^2 > r^2. \quad (4.5)$$

A unit Gaussian gives $\lambda_1(L) \leq 11/16$, and the comparison $t^4/4 \geq (169/100)t^2 - (169/100)^2$ gives $\lambda_5(L) \geq 88439/10000$. By parity, the residual theorem applied to (4.5) isolates a single even eigenvalue in the stated interval, which is therefore $\lambda_3(L)$. $\square$

Corollary 4.3. *At every target,*

$$\Delta - 3a^2 > G(\alpha, a, v) := \frac{467}{125} - \frac{125}{61}\alpha^2 + 2\alpha a - \frac{15}{4}a^2 - \frac{3}{4}v. \quad (4.6)$$

Proof. For $\theta = (4/5)^3 = 64/125$, completing the square gives $t^4/4 - \alpha t^2 \geq \theta t^4/4 - \alpha^2/(1 - \theta)$ with $1 - \theta = 61/125$, so $H_\alpha \geq p^2 + \theta t^4/4 - \frac{64}{61}\alpha^2$. Quartic scaling multiplies the eigenvalues of $p^2 + \theta t^4/4$ by $\theta^{1/3} = 4/5$, so by [Lemma 4.2](#)

$$\lambda_3(\alpha) > \frac{4}{5} \cdot \frac{467}{100} - \frac{64}{61}\alpha^2 = \frac{467}{125} - \frac{64}{61}\alpha^2. \quad (4.7)$$

At a target the virial identity gives $\lambda_1 = \frac{3}{4}(a^2 + v) - 2\alpha a + \alpha^2$. Subtracting it from (4.7) and then subtracting $3a^2$ produces (4.6): the coefficient of α^2 moves from $64/61$ to $64/61 + 1 = 125/61$ because of the $+\alpha^2$ in λ_1 , and the coefficient of a^2 from 3 to $3 + \frac{3}{4} = \frac{15}{4}$. $\square$

4.3. The bad set and its exact constraints. Let $m_j = \langle t^j \rangle$, so $m_0 = 1$, $m_2 = a$ and $m_4 = a^2 + v$. Eigenvalue and hypervirial integration by parts give, for $k \geq 1$,

$$m_{2k+4} = \frac{4(2k+1)}{2k+3}(\lambda_1 - \alpha^2)m_{2k} + \frac{8\alpha(k+1)}{2k+3}m_{2k+2} + \frac{2k(2k-1)(2k+1)}{2k+3}m_{2k-2}, \quad (4.8)$$

which determines $m_6, \dots, m_{20}$ as rational polynomials in (α, a, v) . Beyond the four inequalities of [Lemma 3.2](#) and (3.13), three further constraints are expectations of squares,

$$\langle q_i(t^2)^2 \rangle = \sum_{j,k=0}^5 q_{i,j} q_{i,k} m_{2(j+k)} \geq 0, \quad i = 1, 2, 3, \quad (4.9)$$

with

$$\begin{aligned} 10^5 q_1 &= (-13381, -31687, 100000, -48651, 7725, -379), \\ 10^5 q_2 &= (-15103, -58143, 100000, -32684, 2975, -35), \\ 10^5 q_3 &= (-485, -70487, 100000, -37526, 4978, -207). \end{aligned} \quad (4.10)$$

Lemma 4.4. *Every target lies in the closed rational box*

$$0 \leq \alpha \leq \frac{4}{5}, \quad 0 \leq a \leq \frac{5}{2}, \quad 0 \leq v \leq 2. \quad (4.11)$$

Proof. The first bound is [Proposition 3.6](#). If $a \geq 1/4$, the left side of (3.13) before subtracting v is maximized at $\alpha = 4/5$, is decreasing at and beyond $a = 5/2$, and equals $-8/15$ there; hence $a < 5/2$. The response inequality (3.6) then gives $v^2 < 10/3 < 4$. $\square$

A hypothetical target at which $G \leq 0$ therefore makes nine rational polynomials simultaneously nonnegative on the box (4.11): the four of [Lemma 3.2](#), the Gaussian–virial constraint (3.13), the three squares (4.9), and $-G$. Call the set of such points the *bad set*.

4.4. Finite rational Bernstein exclusion. Map (4.11) onto $[0, 1]^3$ by $\alpha = \frac{4}{5}x$, $a = \frac{5}{2}y$, $v = 2z$. For a power polynomial $p = \sum_{j \leq d} A_j x^{j_1} y^{j_2} z^{j_3}$ the tensor Bernstein coefficients are

$$B_i = \sum_{j \leq i} A_j \prod_{\ell=1}^3 \frac{\binom{i_\ell}{j_\ell}}{\binom{d_\ell}{j_\ell}}, \quad (4.12)$$

and the range of p over the box lies in the convex hull of the B_i ; exact de Casteljau bisection supplies the coefficients of each child box. The traversal always bisects a least-refined coordinate, breaking ties in the order α, a, v , and closes a box only when the greatest Bernstein coefficient of one required polynomial is strictly negative. Carried out in exact rational arithmetic it terminates, with the closures distributed as follows.

Excluding polynomial	Leaves
$3a - 2\alpha$	0
$4a/3 - v^2$	78
Heisenberg–virial	9
Gaussian–virial	14
Robertson–hypervirial	18
$\langle q_1^2 \rangle$	33
$\langle q_2^2 \rangle$	72
$\langle q_3^2 \rangle$	66
$-G$	89

Proposition 4.5. *The bad set is empty: $G > 0$ at every target.*

Proof. The traversal produces a finite rooted binary tree with 379 terminal nodes, 378 internal nodes, and maximum total depth 24. At each terminal node, direct substitution in the rational de Casteljau recurrence gives a strictly negative greatest Bernstein coefficient for the polynomial named in the table, so that polynomial is negative throughout the corresponding closed dyadic box and the box contains no point of the bad set.

For exhaustiveness, label each child by appending L or R to its parent’s binary path. The terminal paths are prefix-free, and their exact dyadic weights satisfy

$$\sum_{\text{terminal paths } w} 2^{-|w|} = 1. \quad (4.13)$$

Since every internal node has exactly the two de Casteljau children of its own box, prefix-freeness together with (4.13) forces the terminal boxes to cover $[0, 1]^3$, boundaries included. Every point of the root cube therefore lies in a terminal box, and every terminal box is excluded. The inputs to this finite calculation are (4.8), (4.9), (4.10), (4.12), the affine map above, and the stated deterministic split rule; no floating-point estimate and no stored polynomial enters. $\square$

Theorem 4.6. *$f'(\alpha) = \lambda'''(\alpha) < 0$ at every target.*

Proof. Proposition 4.5 gives $G > 0$ at every target, Corollary 4.3 then gives $\Delta > 3a^2$, and Lemma 4.1 converts that into $\lambda'''(\alpha) < 0$. $\square$

5. A TARGET EXISTS

Proposition 5.1. *f has at least one positive zero.*

Proof. Suppose not. By Proposition 3.1, $f(0) > 0$, so continuity and the absence of a zero give

$$\lambda''(\alpha) > \frac{4}{3} \quad (\alpha \geq 0), \quad (5.1)$$

and integrating twice,

$$\lambda(\alpha) > \lambda(0) + \lambda'(0)\alpha + \frac{2}{3}\alpha^2 \quad (\alpha > 0). \quad (5.2)$$

Against this stands a variational upper bound. For $\rho > 0$ put $s = \sqrt{2\alpha}$ and take the normalized translated Gaussian

$$\psi_{\rho,s}(t) = \left(\frac{\rho}{\pi}\right)^{1/4} \exp\left[-\frac{\rho}{2}(t-s)^2\right].$$

Writing $X = t - s$ one has $\langle X \rangle = \langle X^3 \rangle = 0$, $\langle X^2 \rangle = 1/(2\rho)$ and $\langle X^4 \rangle = 3/(4\rho^2)$, and $s^2 = 2\alpha$ gives $t^2/2 - \alpha = sX + X^2/2$, so

$$\langle p^2 \rangle = \frac{\rho}{2}, \quad \left\langle \left(\frac{t^2}{2} - \alpha\right)^2 \right\rangle = \frac{\alpha}{\rho} + \frac{3}{16\rho^2}. \quad (5.3)$$

Choosing $\rho = \sqrt{2\alpha}$ in the Rayleigh quotient,

$$\lambda(\alpha) \leq \sqrt{2\alpha} + \frac{3}{32\alpha} = o(\alpha^2), \quad (5.4)$$

which contradicts (5.2) as $\alpha \rightarrow \infty$. $\square$

6. THE TARGET IS UNIQUE

Lemma 6.1. *Let $I \subset \mathbb{R}$ be an interval and $F \in C^1(I)$ with $F'(x) < 0$ at every zero x of F . Then F has at most one zero in I .*

Proof. Suppose $x_1 < x_2$ are zeros. The negative derivatives give $F < 0$ immediately to the right of x_1 and $F > 0$ immediately to the left of x_2 , so there are $p < q$ between them with $F(p) < 0 < F(q)$. Let z be the first zero of F in $[p, q]$. Continuity gives $F < 0$ immediately to the left of z , whereas $F'(z) < 0$ gives $F > 0$ there — a contradiction. $\square$

Proof of Theorem 1.1. By Proposition 2.1 it suffices to show that f has exactly one real zero. Theorem 4.6 gives $f' < 0$ at every zero of f , so Lemma 6.1 leaves at most one; Proposition 5.1 gives at least one. Hence there is exactly one target α_* , and Propositions 3.1 and 3.6 place it in $(0, 4/5)$.

Finally $b \mapsto \alpha = -b/4$ is a bijection and (2.3) gives $m'(b) = 0$ exactly when $f(-b/4) = 0$, so m has the single critical point $b_* = -4\alpha_* \in (-16/5, 0)$, and

$$m''(b_*) = -\frac{3}{8}f'(\alpha_*) > 0$$

by Theorem 4.6, making it a strict local minimum. $\square$

7. SCOPE AND CONCLUSION

Theorem 1.1 concerns the ground eigenvalue and all real b . It does not locate b_* sharply, address excited-state curvature levels, or settle the infinite-mode field problem that motivated the oscillator. The result is logically distinct from uniqueness of a zero of λ'_1 and from the separate positivity and coercivity problem for m .

CODE AVAILABILITY

One step of the proof is computer assisted: Proposition 4.5, the exclusion of the bad set, rests on a finite traversal that was carried out rather than written out. An exact verifier that reproduces it in rational arithmetic accompanies this paper; it rebuilds the nine polynomials from the recurrence (4.8), converts them by (4.12), subdivides the box (4.11), and prints the node, leaf, depth, and per-polynomial counts reported in Section 4.

It is a check and not a premise. The nine polynomials, the affine map, the deterministic split rule, the strict terminal test, and the prefix-free coverage argument are all stated above, so the argument can be followed without running anything, and no floating-point value enters at any point.

ACKNOWLEDGMENTS

Z. Y. acknowledges support from the Strategic Priority Research Program of the Chinese Academy of Sciences under Grant No. XDB1680102. Z. Y. also acknowledges support by National Natural Science Foundation of China (Grant No. 12247101). We also acknowledge Gewu Intelligence Lab for providing the collaborative research environment in which this project was developed.

OpenAI GPT-5.6 Sol was used substantively in this work: in solving the problem, including the formulation of arguments and the derivation of intermediate steps, and in drafting and revising the manuscript. The authors specified the problem, guided the approach to be taken, checked every argument and result it produced, and take full responsibility for the content of this manuscript.

REFERENCES

1. Shmuel Agmon, *Lectures on exponential decay of solutions of second-order elliptic equations: Bounds on eigenfunctions of n -body schrodinger operations. (mn-29)*, Princeton University Press, December 1983.
2. John C Baez and Zhengfang Zhou, *Renormalized oscillator hamiltonians*, Advances in Mathematics **92** (1992), no. 1, 106–127.
3. Hajer Bahouri, Davide Barilari, Isabelle Gallagher, and Matthieu Léautaud, *Spectral summability for the quartic oscillator with applications to the engel group*, Journal of Spectral Theory **13** (2023), no. 2, 623–706.
4. Carl M. Bender and Tai Tsun Wu, *Anharmonic oscillator*, Physical Review **184** (1969), no. 5, 1231–1260.
5. Rida T. Farouki, *The bernstein polynomial basis: A centennial retrospective*, Computer Aided Geometric Design **29** (2012), no. 6, 379–419.
6. Søren Fournais and Bernard Helffer, *Spectral methods in surface superconductivity*, Birkhäuser Boston, 2010.
7. B HELFFER and A MORAME, *Magnetic bottles for the neumann problem: curvature effects in the case of dimension 3 (general case)*, Annales Scientifiques de l'École Normale Supérieure **37** (2004), no. 1, 105–170.
8. B. Helffer and M. Persson, *Spectral properties of higher order anharmonic oscillators*, Journal of Mathematical Sciences **165** (2010), no. 1, 110–126.
9. Bernard Helffer, *The Montgomery model revisited*, Colloquium Mathematicum **118** (2010), no. 2, 391–400.
10. Bernard Helffer and Matthieu Leautaud, *On critical points of eigenvalues of the Montgomery family of quartic oscillators*, Indiana University Mathematics Journal **73** (2024), no. 6, 2073–2133.
11. Bernard Helffer and Abderemane Mohamed, *Semiclassical analysis for the ground state energy of a schrödinger operator with magnetic wells*, Journal of Functional Analysis **138** (1996), no. 1, 40–81.
12. Tosio Kato, *Perturbation theory for linear operators*, Springer Berlin Heidelberg, 1995.
13. Richard Montgomery, *Hearing the zero locus of a magnetic field*, Communications in Mathematical Physics **168** (1995), no. 3, 651–675.
14. Ramon E. Moore, R. Baker Kearfott, and Michael J. Cloud, *Introduction to interval analysis*, Society for Industrial and Applied Mathematics, January 2009.
15. Xing-Bin Pan and Keng-Huat Kwek, *Schrödinger operators with non-degenerately vanishing magnetic fields in bounded domains*, Transactions of the American Mathematical Society **354** (2002), no. 10, 4201–4227.
16. Michael Reed and Barry Simon, *Methods of modern mathematical physics IV: Analysis of operators*, Academic Press, New York, 1978.
17. Siegfried M. Rump, *Verification methods: Rigorous results using floating-point arithmetic*, Acta Numerica **19** (2010), 287–449.
18. Barry Simon, *Coupling constant analyticity for the anharmonic oscillator*, Annals of Physics **58** (1970), no. 1, 76–136, With an appendix by A. Dicke.
19. Alexander V Turbiner and J C del Valle, *Anharmonic oscillator: a solution*, Journal of Physics A: Mathematical and Theoretical **54** (2021), no. 29, 295204.

LANZHOU CENTER FOR THEORETICAL PHYSICS, KEY LABORATORY OF THEORETICAL PHYSICS OF GANSU PROVINCE, KEY LABORATORY OF QUANTUM THEORY AND APPLICATIONS OF MOE, GANSU PROVINCIAL RESEARCH CENTER FOR BASIC DISCIPLINES OF QUANTUM PHYSICS, LANZHOU UNIVERSITY, LANZHOU 730000, CHINA

Email address: yaozy@lzu.edu.cn